Further benchmarking for railway routing and scheduling in Flaspland

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Outline

- Introduction
- Where we left off
- Experiments
- Toolkit
- Conclusion

INTRODUCTION

Flatland

What is Flatland?

Grid-based virtual railway environment for a global R&S competition.



What Flatland considers

- Starting and ending stations
- Collision avoidance
- Different speeds
- Time windows

Flatland

What is Flatland?

Grid-based virtual railway environment for a global R&S competition.



What Flatland does not consider

- Intermediate stations (recent update)
- Train lengths
- Train yards & shunting
- Passengers

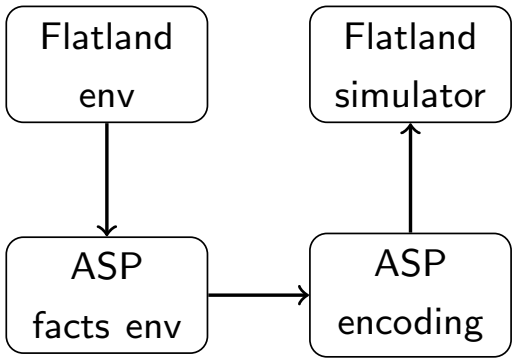
Flaspland

What is Flaspland?

A toolkit that bridges the gap between Flatland and ASP.

Workflow:

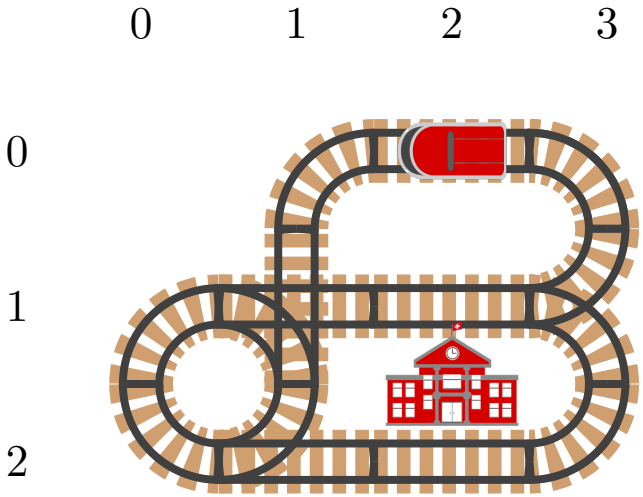
- Generates environments
- Converts environments to ASP facts
- Converts ASP solutions to Flatland actions
- Runs simulation

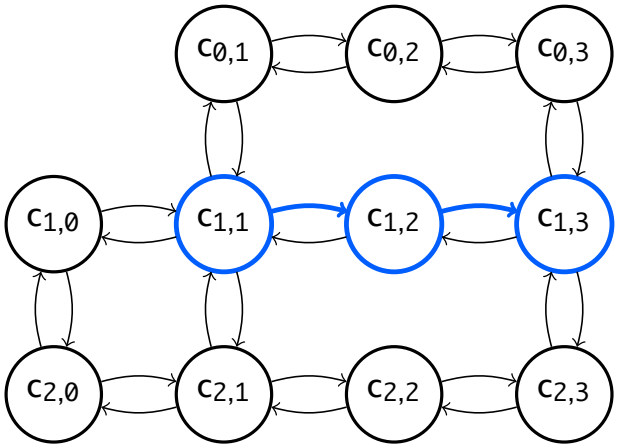


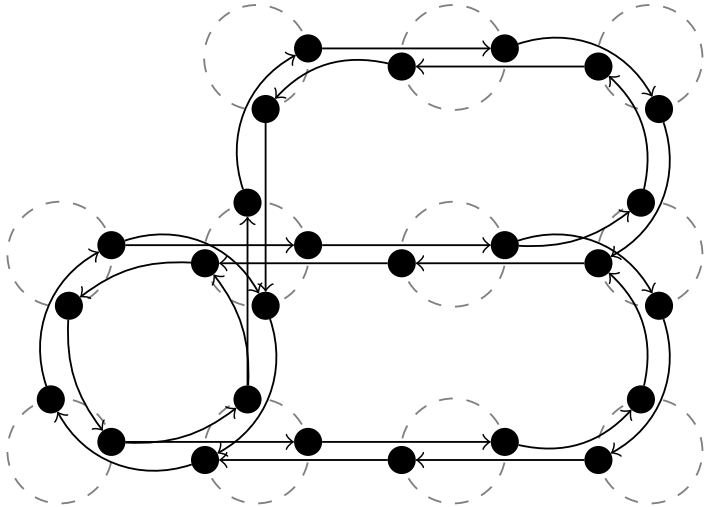
WHERE WE LEFT OFF

Previous discussions

Graph translations:







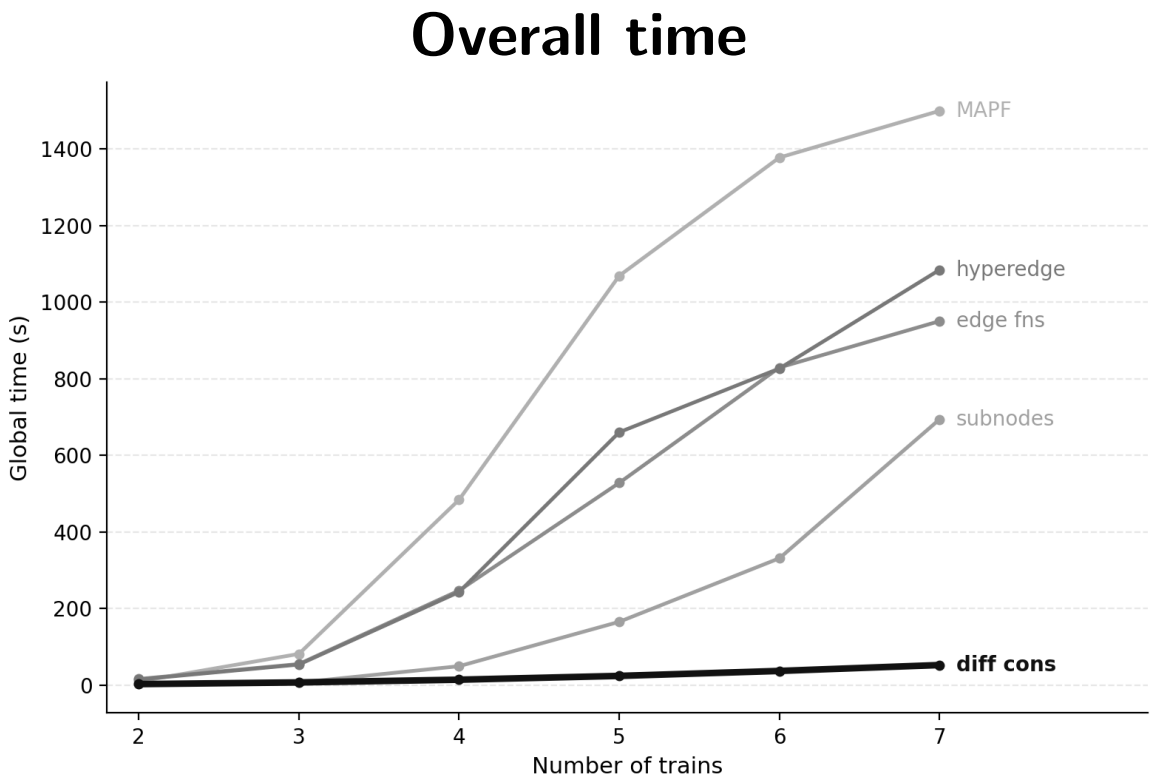
Previous discussions

Three graph translations:

1. EdgFuncs
2. SubNodes
3. HypEdges

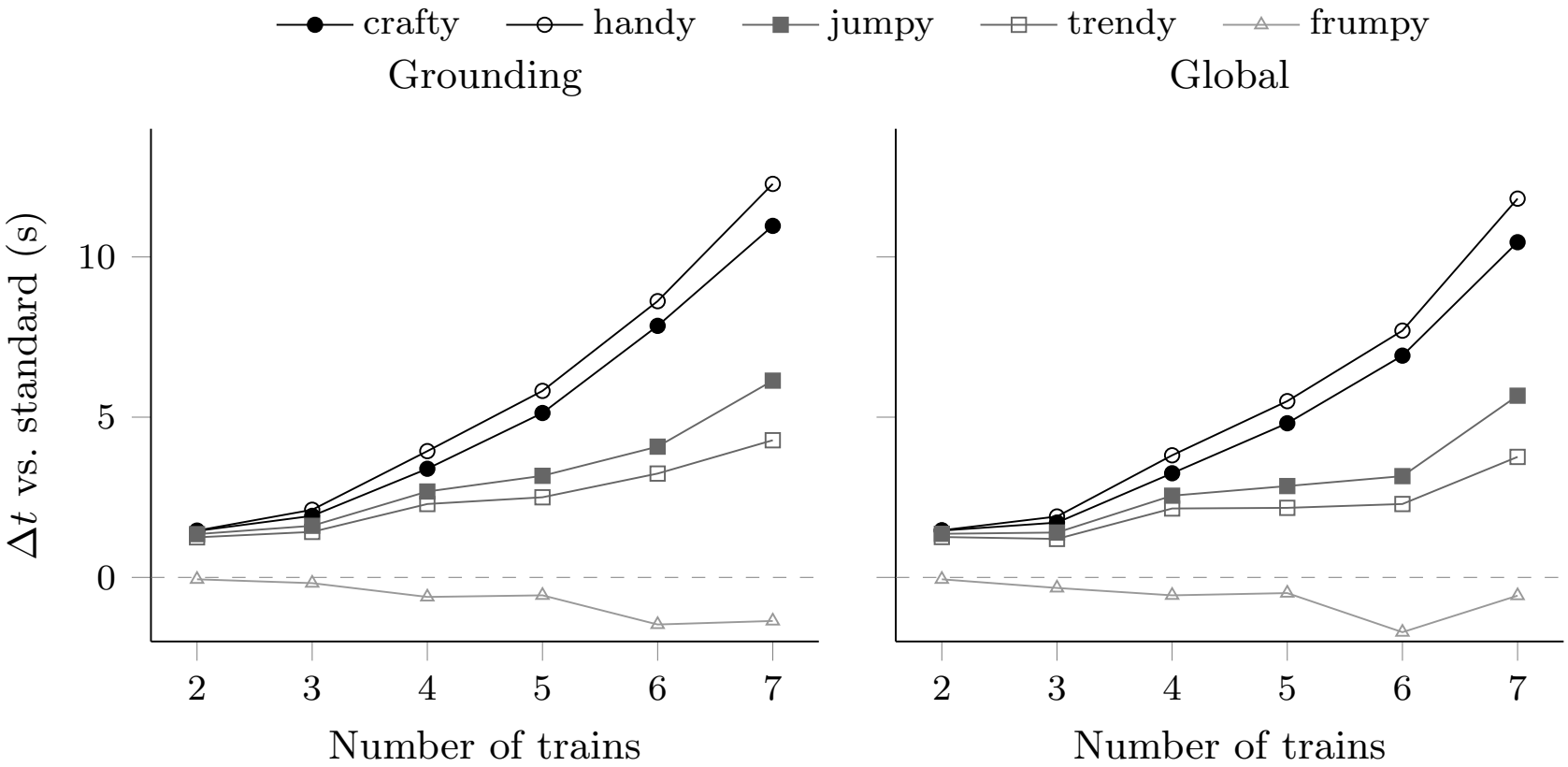
Two approaches:

1. ASP
2. DifCons



EXPERIMENTS

Experiment #1: Configurations



Experiment #2: Graph Representations

Total times:

Encoding	**Graph**	**Avg. total time**
DifCons	SubNodes	30.1
drive-map	SubNodes	248.0
DifCons	HypEdges	340.2
drive-map	HypEdges	743.6
DifCons	EdgFuncs	673.2
drive-map	EdgFuncs	678.7

Experiment #2: Graph Representations

Grounding times:

Encoding	**Graph**	**Avg. grounding time**
drive-map	SubNodes	8.7
DifCons	SubNodes	28.3
drive-map	HypEdges	32.7
DifCons	HypEdges	79.5
drive-map	EdgFuncs	36.5
DifCons	EdgFuncs	93.4

TOOLKIT

Recent improvements

- CLI commands
  - `flaspland build`
  - `flaspland solve`
- Default configuration file
- Error handling
- Documentation

CONCLUSION

Next steps

Research side:

- Scaling up to larger environments
- New encoding techniques
- Optimization

Toolkit improvements:

- Adding malfunctions support to the toolkit
- Support for official Flatland updates

QUESTIONS?